

Iyad Elwy

+49 152 0795 9219 | Stuttgart, Germany | [Online Portfolio — portfolio.iyadelwy.de](#) | elwyiyad@gmail.com | [LinkedIn](#) | [Github](#)

Summary

Enjoy working on projects at the intersection of **data engineering**, **DevOps**, and **infrastructure**. Interests include **container orchestration**, workflow **automation**, and building efficient, **scalable Software Systems**.

Employment History

Software Engineer		Oct 2025 – Current Date
Index Labs – Reutlingen/München, Germany		
- Founding developer establishing technical infrastructure and development processes from ground zero for early-stage startup - Onboarded and mentored new developers, coordinating work and establishing team collaboration practices - Architected complete CI/CD infrastructure using GitLab CI/CD, Docker, and Kubernetes for all company projects - Led requirements engineering and system design, decomposing complex systems into scalable microservices architectures - Built production backend services using Python/FastAPI with PostgreSQL, Redis, and RabbitMQ for data processing - Designed and developed Dave (data hub), Lara (book review system), and Fine (invoice automation) - Integrated enterprise systems including Oxaion ERP and Elasticsearch for automated data flows and process automation - Leveraged AI-assisted development using Claude with advanced prompting: extended thinking, chain-of-thought reasoning, few-shot learning - Restructured monolithic systems into cloud-native architectures for flexible deployment across local, cloud, and K8s - Managed multi-environment deployments (dev, staging, production) with configuration management and security practices - Implemented cost-effective infrastructure solutions, optimizing spending through strategic technology evaluation		
Data Engineer (PreMaster Program)		Nov 2024 – Sep 2025
Bosch GmbH – Renningen, Germany		
- Part of the Data and Tooling team for the Interior Sensing Solutions (ISS) project at Bosch - Managing the lifecycle of Airflow deployments for data workflows - Working with Kubernetes and Rancher to deploy containerized applications - Supporting multinational teams with debugging Kubernetes deployments and resources - Responsible for migrating Kubernetes resources between clusters with different versions - Managing, renewing, and evaluating TLS certificates for deployments and services - Introducing and configuring Spark cluster mode for fast and efficient data processing and KPI calculations - Deploying Databricks Unity Catalog and migrating all tables to the catalog for centralized, governance-compliant data management and access control - Creating and expanding Helm charts for services and tools - Writing and debugging Python scripts to extract, process, and load data into our data lake or data warehouse		
Data Science Internship		Feb 2024 – Jun 2024 (5 Mo)
Bosch GmbH – Bühl, Germany		
- Contributed to the development of predictive time series algorithms to shorten the time-to-market for our product and worked on GenAI use cases - Used visualization and statistical analysis techniques to assess data integrity and uncover valuable insights - Developed machine learning and deep learning models trained on machine sensor data - Used Azure Cloud and Azure OpenAI to modify and test models (LLMs) for performance optimization - Integrated Azure OpenAI with Microsoft Copilot to develop custom services that boosted the team's daily productivity - Leveraged Azure Cognitive Services for natural language processing tasks to improve the extraction of insights		
Education		
M.Sc. Professional Software Engineering Oct 2025 – Sep 2027 Knowledge Foundation X Hochschule Reutlingen – Stuttgart, Germany Schwerpunkt auf Cloud-Native und skalierbare Softwarearchitektur	B.Sc. Informatics & Computer Science Apr 2020 – Jul 2024 German International University – Cairo, Egypt Major Data Science Minor Software Engineering	German International Abitur Apr 2010 – Jun 2019 Europa Schule Kairo – Cairo, Egypt Obtained my German International Abitur from Europa Schule Kairo

Achievements & Publications

- Elwy, Iyad, & Sabty, Caroline. "Takween: A comprehensive Arabic collection and annotation tool." [Procedia Computer Science 244 \(2024\): 186-193.](#)

Professional Skills

Software: Python JavaScript SQL Node.js FastAPI NextJs Git Debian Based Linux Docker Airflow Github Actions Gitlab CI/CD PostgreSQL Redis RabbitMQ MongoDB RESTful APIs Microservices Architectures	Data: Spark Pandas Polars Pytorch Scikit-Learn Data Preprocessing Data-Pipelining Visualization DuckDB Understanding of Deep Neural Networks	Infrastructure: Kubernetes Helm System Design Dev-Ops Workflows Cloud Design Patterns Docker Compose CI/CD Automation Multi-Environment Deployments Container Orchestration GitOps
AI & LLM: Prompt Engineering AI-Assisted Development Claude API	Leadership: Requirements Engineering Mentoring Agile/Scrum System Design German (Native language level C2)	Englisch (Fluent C1)